



TONY BONNAIRE

Research Scientist @CNRS | Paris-Saclay University

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tbonnair

Current research interests

- Theory: Memorization in generative models, physics-inspired diffusion models, non-convex optimization.
- Applications: Generative models for cosmological data and critical phenomena in statistical physics.

EXPERIENCE

Research scientist – CNRS, Paris-Saclay University

Diffusion Models Non-convex optimization AI4Science – Python PyTorch

Oct. 2025 – Ongoing

Paris, France

- Machine learning theory (optimization, score-based generative models),
- Generative Models for physics (cosmology, statistical physics).

AI Research Fellow – ENS Paris, PSL University, PR[AI]RIE

Diffusion Models High-dimensional statistics AI4Science – Python PyTorch

Jan. 2023 – Oct. 2025

Paris, France

- Machine learning theory (optimization, generative models),
- AI for cosmology (segmentation, generative modelling),
- ML teaching to master students of PSL University (150 hrs/year).

Postdoctoral researcher – ENS Paris, PSL University, PR[AI]RIE

High-dimensional statistics Analytical – Python PyTorch

Nov. 2021 – Jan. 2023

Paris, France

- Machine learning theory (high-dimensional non-convex optimization).

Ph.D. Student – IAS, Paris-Saclay University

AI4Science Cosmology Pattern extraction – Python

Oct. 2018 – Sept. 2021

Orsay, France

- 2D/3D pattern extraction from cosmological data and simulations.
- Statistical analysis of the cosmic web to improve cosmological models.
- Supervision: Nabila Aghanim, Aurélien Decelle.

Research engineer – Thales

Algorithmics Pattern extraction – C Matlab OpenCL

Sept. 2017 – Sept. 2018

Élancourt, France

- Design and implementation of (un)supervised algorithms for electronic warfare problems.

EDUCATION

Ph.D. in Astronomy and Astrophysics

Paris-Saclay University

Oct. 2018 – Nov. 2021

Orsay, France

CentraleSupélec Engineer

CentraleSupélec, track: Applied mathematics (rank: top 10%)

Sept. 2014 – Sept. 2017

Gif-sur-Yvette, France

LANGUAGES

French



English



Italian



PROGRAMMING

Python PyTorch Git
Slurm C R Matlab
OpenCL Objective-C

REFERENCES

Prof. Giulio Biroli

@ LPENS

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Laboratoire de Physique de l'École normale supérieure, ENS, Paris, France.

Dr. Nabila Aghanim

@ IAS

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Institut d'Astrophysique Spatiale, Université Paris-Saclay, Orsay, France.

Dr. Aurélien Decelle

@ UCM

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Departamento de Física Teórica I, Universidad Complutense, Madrid, Spain.

TEACHING

Machine Learning for Scientific Discovery

ENS-PSL Co-advisor Co-lecturer Sept. 2025 – Ongoing
36 hours – *Unified Machine Learning* introductory course proposed to students of ENS together with Prof. Marc Lelarge.

Machine Learning Principles with Applications in Physics

Physics Department, Master ICFP, ENS-PSL Advisor Co-lecturer Sept. 2023 – Ongoing
64 hours – *Machine learning* introduction for M1 students, and supervision of tutored projects.

Artificial Intelligence and Chemistry

Chemistry Department, ENS-PSL Advisor Lecturer Sept. 2023 – Ongoing
32 hours – *Introduction to the basics of machine learning*, and supervision of tutored projects.

PSL AI Hackathons

ENS-PSL, Dauphine Advisor Instructor Jan. 2023 – Ongoing
50 hours – *Preparation, organization, supervision, and participation in final juries of the PSL AI hackathons for master students.*

Data Challenges

Master MVA (Paris-Saclay University) Nov. 2022 – Jun. 2025
Organisation and beta-testing of the data science challenges for the web platform "Challenge Data ENS".
Evaluation of the M2 students from the MVA Master course of Prof. Stéphane Mallat.

SUPERVISION

- Mehdi Noor – Co-supervision of Ph.D. Sept. 2024 – Ongoing
Emulating the Cosmic Web with Generative AI: Methods and Applications
- Mehdi Noor – Co-supervision of M2 internship Mar. 2024 – Sep. 2024
Generation of Cosmological Simulations via Diffusion-based Models

PROJECTS

- 2024 – Ongoing: Membership to the Euclid consortium (Galaxy/AGN Evolution Working Group).
- 2024 – Ongoing: Collaborator of an ERC Proof Of Concept (PoC) for the transfer of methodologies developed during my Ph.D. in cosmology to health science with Dr. N. Aghanim and Dr. A. Decelle.

PRIZES & GRANTS

- 2025: Recipient of one of the four best paper awards at NeurIPS 2025.
- 2025: Allocation of 46,000 GPU hours on Jean-Zay for research on theory of diffusion models (AD011016319).
- 2023: 5-year fellowship at PSL University and ENS Paris for research and teaching in Artificial Intelligence.
- 2022: Prix de la Chancellerie des Université de Paris, catégorie Sciences.

COMMUNITY SERVICES AND SHARING

- Codes
Tree-based Ridge eXtractor (T-ReX): A publicly-available Python implementation of the filament-finder method.
- Reviewing activities
Machine Learning conferences: NeurIPS 2024 Scientific Methods for Understanding Deep Learning workshop.
Physics journals: MNRAS, Nature Scientific Reports, JSTAT.

RELEVANT PUBLICATIONS

Ongoing works and preprints

- Giulio Biroli, **Tony Bonnaire**, Gilles Tarjus, and Dariel Vllamasi. "Stochastic Interpolants, Diffusion Models and Exact Renormalization Group Methods". 2026.
- Tony Bonnaire** and Giulio Biroli. "A replica analysis of the asymptotic fluctuations of extreme eigenvalues in non-white Wishart matrices". 2025.

Machine learning: theory, methods, and applications

- [1] **Tony Bonnaire**, Giovanni Catania, Aurélien Decelle, and Beatriz Seoane. “On the role of non-linear latent features in bipartite generative neural networks”. In: **SciPost Phys.** 19 (Dec. 2025), p. 141. DOI: 10.21468/SciPostPhys.19.6.141.
- [2] **Tony Bonnaire**, Raphaël Urfin, Giulio Biroli, and Marc Mézard. “Why Diffusion Models Don’t Memorize: The Role of Implicit Dynamical Regularization in Training”. In: **NeurIPS 2025 (oral)** (Mar. 2025). arXiv: 2505.17638 [cs.LG].
- [3] **Tony Bonnaire**, Giulio Biroli, and Chiara Cammarotta. “The Role of the Time-Dependent Hessian in High-Dimensional Optimization”. In: **J. Stat. Mech.** (Feb. 2025).
- [4] Giulio Biroli, **Tony Bonnaire**, Valentin De Bortoli, and Marc Mézard. “Dynamical regimes of diffusion models”. In: **Nature Comm.** 15.1 (Nov. 2024), p. 9957.
- [5] **Tony Bonnaire** et al. “High-dimensional non-convex landscapes and gradient descent dynamics”. In: **J. Stat. Mech.** 2024.10 (Oct. 2024), p. 104004. DOI: 10.1088/1742-5468/ad2929.
- [6] Victor Bonjean et al. “Self-supervised component separation for the extragalactic submillimetre sky”. In: **Astron. Astrophys.** 686, A91 (June 2024), A91. DOI: 10.1051/0004-6361/202245624.
- [7] **Tony Bonnaire**, Aurélien Decelle, and Nabila Aghanim. “Regularisation of Mixture Models for Robust Principal Graph Learning”. In: **IEEE Trans. Pattern Anal.** 44 (Dec. 2021), pp. 9119–9130. DOI: 10.1109/TPAMI.2021.3124973.
- [8] **Tony Bonnaire**, Aurélien Decelle, and Nabila Aghanim. “Cascade of phase transitions for multiscale clustering”. In: **Phys. Rev. E** 103 (Jan. 2021), p. 012105. DOI: 10.1103/PhysRevE.103.012105.
- [9] **Tony Bonnaire**, N. Aghanim, A. Decelle, and M. Douspis. “T-ReX: a graph-based filament detection method”. In: **Astron. Astrophys.** 637 (Sept. 2020), A18. DOI: 10.1051/0004-6361/201936859.

Statistical analyses for cosmology

- [1] Euclid Collaboration et al. “Euclid Quick Data Release (Q1). The role of cosmic connectivity in shaping galaxy clusters”. In: **Submitted to Astron. Astrophys.** (2025). eprint: 2503.15332.
- [2] Stefano Gallo et al. “Tracing gaseous filaments connected to galaxy clusters: The case study of Abell 2744”. In: **Astron. Astrophys.** 692 (Dec. 2024), A200.
- [3] Nabila Aghanim et al. “Dissecting a miniature universe: A multi-wavelength view of galaxy quenching in the Shapley supercluster”. In: **Astron. Astrophys.** 689 (Sept. 2024), A332. DOI: 10.1051/0004-6361/202348672.
- [4] **Tony Bonnaire**, Joseph Kuruvilla, Nabila Aghanim, and Aurélien Decelle. “Cosmology with cosmic web environments II. Redshift-space and cross power spectra”. In: **Astron. Astrophys.** 674 (June 2023), A150. DOI: 10.1051/0004-6361/202245626.
- [5] **Tony Bonnaire**, Nabila Aghanim, Joseph Kuruvilla, and Aurélien Decelle. “Cosmology with cosmic web environments I. Real-space power spectra”. In: **Astron. Astrophys.** 651 (May 2022), A146. DOI: 10.1051/0004-6361/202142852.
- [6] C. Gouin, **Tony Bonnaire**, and N. Aghanim. “Shape and connectivity of groups and clusters: Effect of the dynamical state and accretion history”. In: **Astron. Astrophys.** 651 (July 2021), A56. DOI: 10.1051/0004-6361/202140327.

INVITED SEMINARS

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| • Madrid University, Dept. of Theoretical Physics
The Generative Dynamics of Diffusion Models in Large Dimensions | Madrid, Spain, Mar. 2025 |
| • LISN, Paris-Saclay University
The Generative Dynamics of Optimally-Trained Diffusion Models in Large Dimensions | Orsay, France, Dec. 2024 |
| • Department of Astrophysics, École Normale Supérieure
Dynamical Regimes of Diffusion Models | Paris, France, Apr. 2024 |
| • Department of Astrophysics, École Normale Supérieure
The cosmological information of the cosmic web | Paris, France, Feb. 2024 |
| • Centre de Recherche en Informatique, Signal, et Automatique de Lille
The cosmic web: from identification to cosmological parameters | Lille, France, Jun. 2023 |
| • Institut de Physique Théorique, cosmology group
Identifying and quantifying information of the cosmic web | Saclay, France, Jun. 2022 |
| • University of Geneva, cosmology group
Cosmic web environments: from identification to cosmological parameters | Geneva, Switzerland, Dec. 2021 |
| • Laboratoire d'Astrophysique de Marseille
The cosmic web: filaments identification and quantification of the cosmological information | Orsay, France, Dec. 2021 |
| • Institut Elie Cartan, Dept. of Probability and Statistics
Pattern extraction from point-cloud datasets and cosmological applications | Nancy, France, Dec. 2021 |

- Madrid University, Dept. of Theoretical Physics
The principal graph of the cosmic web: learning patterns in point-cloud datasets

Madrid, Spain, Nov. 2021

CONFERENCES & WORKSHOPS

Invited talks

- **Indo-French workshop on *Navigating rugged landscapes*** Bengaluru, India, Nov. 2025
The Role of the Time-Dependent Hessian in High-Dimensional Optimization
- **Collège de France, chaire of Prof. S. Mallat** Paris, France, Feb. 2025
Dynamique des Modèles de Diffusion
- **Workshop on Generative Models** Paris, France, Feb. 2025
The High-Dimensional Generative Dynamics of Diffusion Models
- **AstroParticle workshop** Orsay, France, Nov. 2024
Cosmology with cosmic web environments
- **Physics Informed Machine Learning workshop** Los Alamos, USA, Oct. 2024
The Generative Dynamics of Diffusion Models in Large Dimensions

Contributed Talks

- **NeurIPS2025 (Oral)** San Diego, USA, Dec. 2025
Why Diffusion Models Don't Memorize: The Role of Implicit Dynamical Regularization in Training
- **Statphys29 (Statistical physics of ML and information)** Florence, Italy, Jul. 2025
The Generative Dynamics of Optimally-Trained Diffusion Models in Large Dimensions
- **New Strategies For Extracting Cosmology From Future Galaxy Surveys** Sexten, Italy, Jul. 2023
Improving cosmological constraints using comic web environments
- **Cosmic cartography 2022** (Virtual) Kashiwa, Japan, Mar. 2022
Cosmology with cosmic web environments
- **Sixteenth Marcel Grossmann Meeting** Virtual, Jul. 2021
Constraining cosmological parameters with cosmic environments
- **Elbereth conference** (Virtual) Paris, France, Feb. 2021
Learning the principal graph of the galaxy distribution
- **COSPAR 2021 Workshop on Machine Learning for Space Sciences** Sydney, Australia, Jan. 2021
Learning the principal graph of the galaxy distribution
- **Workshop Orsay-Tartu** Virtual, Jun. 2020
Detection of cosmic filaments using galaxy distribution
- **Workshop "The Cosmic Web in the Local Universe"** Leiden, Netherlands, Jan. 2020
Detecting cosmic filaments from halo distribution
- **Workshop ByoPiC ERC** Hossegor, France, Jun. 2019
Automatic detection of Cosmic Web elements, a review
- **Elbereth conference** Paris, France, Jan. 2019
Uncovering cosmic filaments from galaxy distribution